Computational Text Analysis
IDHEAP Doctoral School

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University of Lausanne · Day 1

Unil.

A Typical European Commission Press Conference

What is it about in one word or sentence?

https://youtu.be/5CZRN0sbUVs

A Typical European Commission Press Conference

- Press conferences show what kind of questions we can ask using computational text analysis:
  - What topics/policy fields do politicians speak about to the public?
  - What goals do governments associate with a policy adopted?
  - How do politicians frame certain issues?
  - What rhetorical devices are used to advertise a policy?
  - Which other actors are credited for a successful policy adoption?

A Typical European Commission Press Conference

- That is why I have analysed all European Commission communications of the European Commission since 1985 in a recent paper (currently under review).
- My research interest: Education and skills policy
- Question: What does the EU want to achieve with its education and skills? What is the goal and has that changed over time?

A Typical European Commission Press Conference

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- My research interest: Education and skills policy
- Question: What does the EU want to achieve with its education and skills? What is the goal and has that changed over time?
- The two contenders discussed in the literature:
  - Education can be social policy (integrating vulnerable groups into society and the labour market) or
  - it can be economic policy (providing human capital for firms).

What can we learn from press conferences?

- This is just one question that can be answered with the dataset I used.
- What other questions could you ask?

What kind of seminar this will be

- Learning…
  - … the foundations you need to apply all kinds of methods of computational text analysis and to understand CTA papers
  - … simple methods that you can directly work with tomorrow
  - … a workflow that serves for all kind of CTA methods
  - … by making mistakes, trying again and then succeeding
  - … by creating a mini-project tomorrow that you walk away with from the course

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  - … by making mistakes, trying again and then succeeding
  - … by creating a mini-project tomorrow that you walk away with from the course
- The newest methods change quickly at the moment. The basics stay the same.
- The biggest barrier for learning CTA is never having done an analysis.

Who am I?

- Postdoc here at IDHEAP in the Social Policy Unit, working on education- and social policy
- PhD at the European University Institute in Florence
- Worked in politics (e.g. writing speeches) and policy (writing government publications and inputs for legislation).
- Now I analyse the type of texts I produced before to better understand how education systems and welfare states in Europe are changing.

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- Now I analyse the type of texts I produced before to better understand how education systems and welfare states in Europe are changing.
- My expectations:
  - To help you to learn a new and useful method
  - That you stick with me for these two days even if some learning processes will seem tough for those new to R and CTA but boring for those with more experience.

Who is this seminar for?

- Those that use text analysis but have not yet used computational methods or R to do it
- Those who have used R / quanteda but want to revisit the basics
- Those who want to learn how to research public policy and public administration with texts
- Those who have no background in neither topic nor method but want to learn
- You are all welcome!

And now you: Who are you and what are your expectations for this seminar?

- Your name
- Your discipline and PhD project
- Whether you have ever used computational text analysis
- What you would like to learn during the seminar

Seminar Overview

Day 1 Foundations & first analyses

01 Finding and selecting the right texts for analysis / Texts as Traces of Political Conflict and Policy Change

02 A Newcomer's Guide to Computational Text Analysis

03 Working with R: Introduction to R & Quanteda

04 Workflow & Descriptive Analyses

05 Word Counts and Dictionaries

06 Selecting Text for Your Mini-Project

Seminar Overview

Day 2 Text-as-data methods & application

01 Word Frequencies: How Simple Methods Can Trump Complex Ones (Martin 2018)

02 Supervised and Unsupervised Machine Learning: Focus on Topic Modelling

03 In-Class Project: Applied Computational Text Analysis

04 Pitching your projects

Practicalities

- 2 days. 09:30-17:30
  - Lunch at around 12:30 & regular short breaks
  - Let me know when you need a break
- The seminar is for you, so do let me know if I can help you with anything
- You do some independent data analysis on the afternoon of day 2. Everything before will prepare you for that small project.
  - The idea: You apply the methods we have learned about directly to whatever question you find interesting.

Practicalities

- Are there any formalities that you need to fulfill for your doctoral programme?
- Generative AI / Claude / ChatGPT:
  - Is heavily used in coding and computational text analysis.
  - For this seminar: Try to do all coding exercises on your own. Only when stuck and asking your neighbors or me did not help, use Gen AI.
    - **We learn about the foundations. It is important to understand these before you augment your coding**
  - Try to understand what you are doing / never follow GenAI blindly

Credits

Some of the materials used in the next sections are based material that the person who first introduced me to computational text analysis shared with me with permission to use. Many thanks to Theresa Gessler!

Other materials are created just for you. Feedback on the material is very welcome after the seminar.

Questions before we start?

01

Texts as Traces of Political Conflict and Change

What kinds of texts do we work with?

- **Produced for research** (open survey answers, interviews) vs. **“found” texts** produced for other purposes (= most of what we use)
- **Politics** texts: speeches, debates, press releases, manifestos, social media posts
- **Policy** texts: laws, communications, reports, descriptions of programmes
- Further distinctions that matter:
  - Spoken vs. written · elite vs. mass
- Each type is a trace of a **different process** and leaves out different things

Grimmer, Roberts & Stewart (2022), Text as Data, Part II

Texts as traces: an agnostic view

"We are not studying the text itself, but rather the political, economic, and social processes that are reflected in it." (Grimmer et al 2022)

- A text is **one version of reality**: written by someone, for an audience, with a purpose
- There is no "true" model and no "right" way to organise a text we analyse
  - The same press conference texts can be sorted ("classified") by tone, by topic or by sub-topic. Which one is useful depends on the question
- So the **question** drives the corpus, the method and the validation

Grimmer, Roberts & Stewart (2022), ch. 2.5 and p. 35

General Principle in Text Analysis: The question defines the corpus

- **Population of interest:** whose texts, which period, which arena?
- **Quantity of interest:** what do you want to measure? Topics, positions, tone …
- The same corpus can be good or bad, depending on the question:
  - Twitter to measure US public opinion on climate change ✕
  - Twitter to study how politicians interact with journalists ✓
- Even when finding the right corpus (= body of text), there are a few things to look out for

Grimmer, Roberts & Stewart (2022)

Four biases to check (I): Who gets to produce texts?

1 Resource bias

Who can afford to produce and store texts?

Actors with money, staff and archives are over-represented.

Example: some agencies cannot release records because they were never scanned.

Grimmer, Roberts & Stewart (2022), ch. 4.2.1–4.2.2

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2 Incentive bias

Who wants a text to exist or to disappear?

Texts are performative. Actors polish, hide, delete or censor what they write. Transparency rules change what gets written down.

Example: the Chinese government removes posts in social media about collective action (King, Pan & Roberts 2013).

Grimmer, Roberts & Stewart (2022), ch. 4.2.1–4.2.2

Four biases to check (II): How do we get the texts?

3 Medium bias

How does the format shape the content?

Character limits, genres and platforms change what is said. Keep in mind: transcripts lose tone and body language compared to video data.

Example: until 2017 a tweet had 140 characters. This allows for far more content in Chinese than in English. There is a language bias through character limits.

Grimmer, Roberts & Stewart (2022), ch. 4.2.3–4.4

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4 Retrieval bias

How did you select the texts?

Keyword searches or topic modelling for text selection select systematically

Example: searching for titles that include the word “ghost” in a list of books finds Hungry Ghosts (a famine history) and misses A Christmas Carol.

Grimmer, Roberts & Stewart (2022), ch. 4.2.3–4.4

Example: UN General Debate speeches

Question (e.g.): How do states position themselves on development, security or climate?

The corpus

Annual General Debate statements at the UN General Assembly

7,507
speeches

199 countries
1970–2015

Checking the four biases

Incentive: speeches address a world audience. They show stated positions, not actual foreign policy

Resource / time: membership grows and changes (decolonisation, end of the USSR); changes in staff and budget of UN (could this affect topics?)

Medium: many speeches were given in other languages. Research analyses English versions introducing a bias

Retrieval: selecting speeches by keyword (e.g. “climate”) misses those that discuss it in other words

UN General Debate Corpus: Baturo, Dasandi & Mikhaylov (2017), Research & Politics

Selection is iterative: How a text corpus becomes a new question

Example: King, Pan & Roberts on censorship in China

1 Collect

Millions of posts from Chinese social media sites. Intention: to test a text analysis method

2 Surprising finding

Going back to the original pages, many posts are gone: they were censored / deleted

3 Re-select

New question: what does the government censor?

Result: posts about collective action are censored, whether or not they criticise the government.

King, Pan & Roberts (2013, 2014); Grimmer, Roberts & Stewart (2022), ch. 2.6 and 4

Texts as representation of political conflict or policy change

- No ‘blind’ methods seminar but applied to public policy and public administration.
- We want to use the methods learned to understand how policies and administrations are changing and why they are changing.
- Before we look at texts let’s clarify important terms: What is public policy and what is politics?

(Re-)visiting the policy process: What is public policy?

- “Public policy is what public officials within government, and by extension the citizens they represent, choose to do or not to do about public problems.” (Kraft & Furlong 2019, p.3)
    - Analysing policy change, we focus on changes in public policy
- Politics is “the interaction between actors involved” (Hassel & Wegrich 2022, p.14) or “the political games being played between the various actors that are, or want to be, involved” (Hassel & Wegrich 2022, p.13) in policy-making.
    - Analysing political conflict we are interested in politics

Textual manifestation of political conflict

- How do political actors interact with each other in the policy process?
  - Express their positions and critique the positions of others via different media channels
  - Hold speeches defending positions or attacking the positions of others
  - Write manifestos and other textual documents to state their position
  - Mass movements chant their position on manifestations and write them on banners.
- Text is therefore not only at the center of politics but also the best way to learn about the positions of the different actors involved and the conflicts between them.

Example: UN Security Council Meeting 23 February 2022 (S/PV.8974)

The President (*spoke in Russian*): The representative of Ukraine has asked for the floor to make a further statement; I give him the floor.

Mr. Kyslytsya (Ukraine): As I said, Sir, relinquish your duties as President of the Council; call President Putin and call Foreign Minister Lavrov to stop the aggression.

I welcome the decision of some members of the Council to meet as soon as possible to consider the necessary decision that would condemn the aggression that the Russian Federation has launched against my people.

There is no purgatory for war criminals; they go straight to hell.

The President (*spoke in Russian*): I shall now make a further statement in my capacity as the representative of the Russian Federation.

I wanted to say, by way of conclusion, that we did not launch an aggression against the Ukrainian people, but against the junta that is in power in Kyiv.

Textual manifestations of policy change

- Texts do not only capture politics (actor interactions) but also policy (result of interactions).
- Important types of documents include e.g. laws, government publications, policy descriptions, government organisational charts, budgets
- Questions to ask: What is changing? What problem does this address?
    - Beyond change, we could also ask statically: What is the policy? Or specifically: How many organisational units does each agency analysed have according to its organisational chart?

Example: European Skills Agenda Communication
What are the problems this aims to address? What is done?

Action 5: Rolling out the European Universities initiative and upskilling scientists

To roll out the European Universities, the Commission, in close cooperation with the stakeholders and the Member States, will:

- *engage in the full rollout of the European Universities initiative under the Erasmus programme (2021—2027) and Horizon Europe, including by removing obstacles to effective and deeper transnational cooperation between higher education institutions and deepening the cooperation with economic operators, in particular to foster the twin transitions. European Universities will set standards for the transformation of higher education institutions across the European Education Area and the European Research Area, also making lifelong learning and talent circulation a reality.*
- *explore options stemming from their research and innovation dimension to help remove obstacles to effective transnational cooperation between higher education institutions, drawing on the lessons learnt during the pilot calls under Erasmus+ and Horizon 2020. The Commission will identify areas of support for Member State action, explore a concrete approach for a "European degree" and the feasibility of a European University statute (to tackle cross-border legal issues) and for a European Recognition and Quality Assurance System.*

Group Work 1: Texts in your field

5 min on your own · 10 min in small groups · short presentation per group

1. Which texts do others in your discipline or field of research analyse?
2. How are they analysed? Which methods (qualitative, quantitative, mixed …)?
3. Which texts are not analysed (sufficiently) yet but could be interesting?

02

A Newcomer’s Guide to Computational Text Analysis

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A Newcomer’s Guide to Computational Text Analysis

01 Why computational text analysis?

02 What is it?

03 Key terms

04 Six principles of text analysis

05 Workflow and choice of method

Why Computational Text Analysis?

- Political conflict, policy change and other social processes we are interested in manifest in text (see last slides)
    - **What other processes can you think of that are represented in text?**
- More and more text is accessible: debates, laws, press releases, social media …
    - … far too much to read and qualitatively analyse

Grimmer, Roberts & Stewart (2022), Text as Data, ch. 1, pp. 3–4

Why Computational Text Analysis?

- Why now? The internet made texts easy to get, laptops do what once needed servers, and computer science developed new tools and models.
- Computational text analysis lets us collect, analyse and systematise large amounts of text
    - and learn about political processes, policy development and the concepts used in policy-making

Grimmer, Roberts & Stewart (2022), Text as Data, ch. 1, pp. 3–4

What is Computational Text Analysis?

The method

"A variant of content analysis that is expressly quantitative, not just in terms of representing textual content numerically but also in analyzing it, typically using computers"

– Kenneth Benoit

The goal

"The use of texts and language to make inferences about human behavior"

– Grimmer, Roberts & Stewart (2022, p. 3)

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- Also known as **"text as data"** or **Natural Language Processing (NLP)**
- In social science the text is **evidence about people and processes**, not the object of interest itself
- Very different from computer science, which mostly cares about search, recommendation and benchmark tasks

Basic Assumptions

1 Text is a trace

A text is an observable implication of an underlying characteristic of interest: its topic, or an attribute of its author such as a political position.

Grimmer, Roberts & Stewart (2022), Text as Data, ch. 1, p. 11

Basic Assumptions

1 Text is a trace

A text is an observable implication of an underlying characteristic of interest: its topic, or an attribute of its author such as a political position.

2 Text can be represented

We represent a text by extracting its features (usually words). We assume the representation entails the meaning of interest.

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Basic Assumptions

1 Text is a trace

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2 Text can be represented

We represent a text by extracting its features (usually words). We assume the representation entails the meaning of interest.

3 Text characteristics can be measured

Analysing the frequency (or embedding) of features with quantitative methods lets us measure the characteristic behind the text.

Keep in mind: a representation is a useful and incomplete summary of a text, not its one "true" structure. Grimmer, Roberts & Stewart call this a **radically agnostic** view.

Grimmer, Roberts & Stewart (2022), Text as Data, ch. 1, p. 11

4 Key terms: What do these mean? What is an example of it?

Corpus

Document

Token

Feature

4 Key terms: From texts to numbers

Corpus

A collection of documents, selected for a research question

e.g. all UN General Debate speeches, 1970–2015

Document

The unit of analysis: a speech, a manifesto, a tweet, a law or a paragraph of one

e.g. the speech of one country at the UN General Assembly in one year

Token

One occurrence of a word after we split a text into pieces

“we the peoples of the United Nations” = 6 tokens

Feature

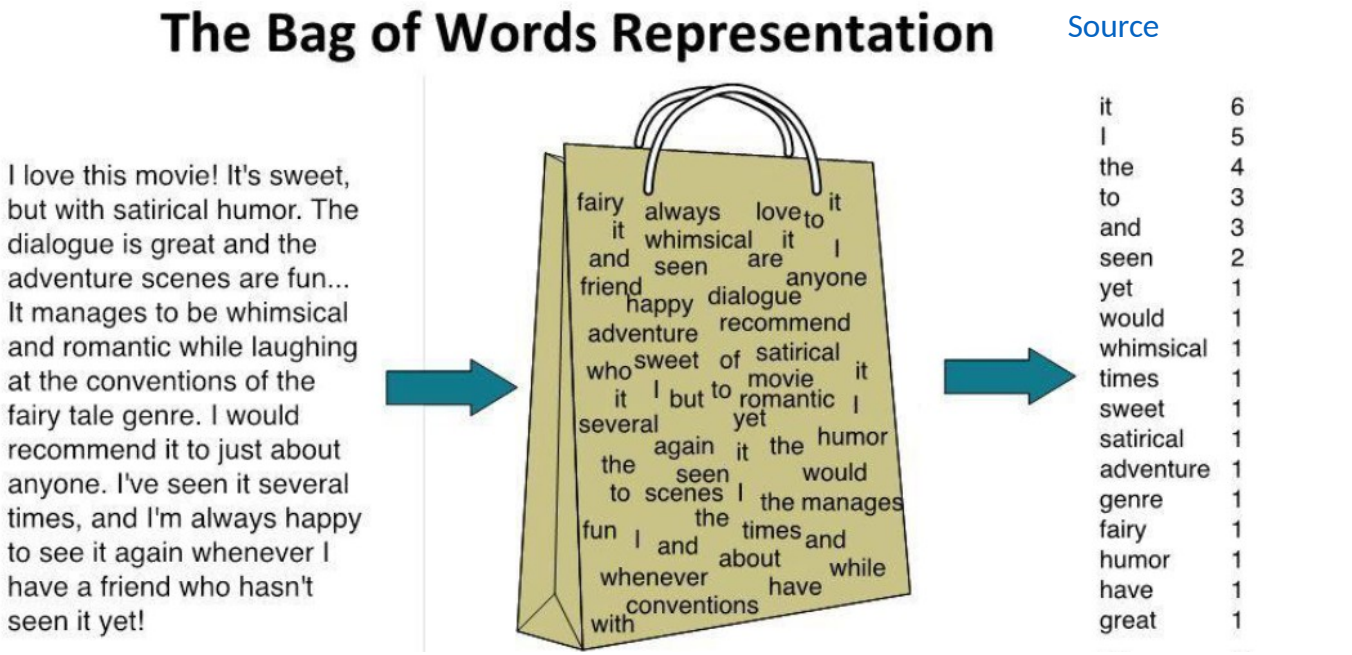
A unique token we treat as a variable. Usually a word, sometimes a phrase

e.g. “climate”, “nations”, “climate change”

The Representation of Text in Computational Text Analysis

- Most of CTA builds on one of two ways to represent (understand and process) text:
  - The bag-of-words representation
  - Word embeddings

The Bag of Words Representation



Document → Tokens → Document-Feature-Matrix

Different methods build on different representations of text

Bag of words

- Each document becomes a list of word counts; word order is not considered
- Reduces complexity, meaning that it uses less information, but works well for many tasks: A useful default
- Our focus during this seminar

e.g. dictionaries, LDA and STM topic models, scaling

Different methods build on different representations of text

No one right way: choose the simplest representation that still captures what you want to measure

Bag of words

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Word embeddings

- In a bag of words, “cat” and “kitten” are as different as “cat” and “table”
- Embeddings are learned from huge outside corpora (Wikipedia, Common Crawl) and place words with similar meanings close together

e.g. BERT Topic Models

Let's get practical

- From now on whenever I explain something theoretical, we will then apply it in R
- I first show it to you on a slide and then we work with R scripts that I made for you
    - Please do not start to work in R while I am still showing you the steps here. You will have time to do it afterwards.
- Now: From document to document-feature-matrix

How does this work? Creating a bag of words

- We need a corpus, which is a collection of texts.
- We tokenize the corpus, so each word represents a feature

```
toks <- tokens(c("This is a corpus for our seminar on computational text analysis and the politics of education",
"This is the second document of this example corpus"))
toks
```

```
## Tokens consisting of 2 documents.
## text1 :
##  [1] "This"          "is"            "a"             "corpus"
##  [5] "for"           "our"           "seminar"       "on"
##  [9] "computational" "text"          "analysis"      "and"
## [ ... and 4 more ]
##
## text2 :
## [1] "This"     "is"       "the"      "second"   "document" "of"       "this"
## [8] "example"  "corpus"
```

Getting word frequencies per document: The document feature matrix

```
dfm(toks)
```

```
## Document-feature matrix of: 2 documents, 19 features (36.84% sparse) and 0 docvars.
##        features
## docs    this is a corpus for our seminar on computational text
##   text1    1  1 1      1   1   1       1  1             1    1
##   text2    2  1 0      1   0   0       0  0             0    0
## [ reached max_nfeat ... 9 more features ]
```

Getting the sorted frequency of features

```
dfm(toks) %>% topfeatures()
```

```
##    this      is  corpus     the      of       a     for     our seminar      on
##       3       2       2       2       2       1       1       1       1       1
```

Six principles of text analysis by Grimmer et al.

1 Theory matters

2 Humans stay in charge

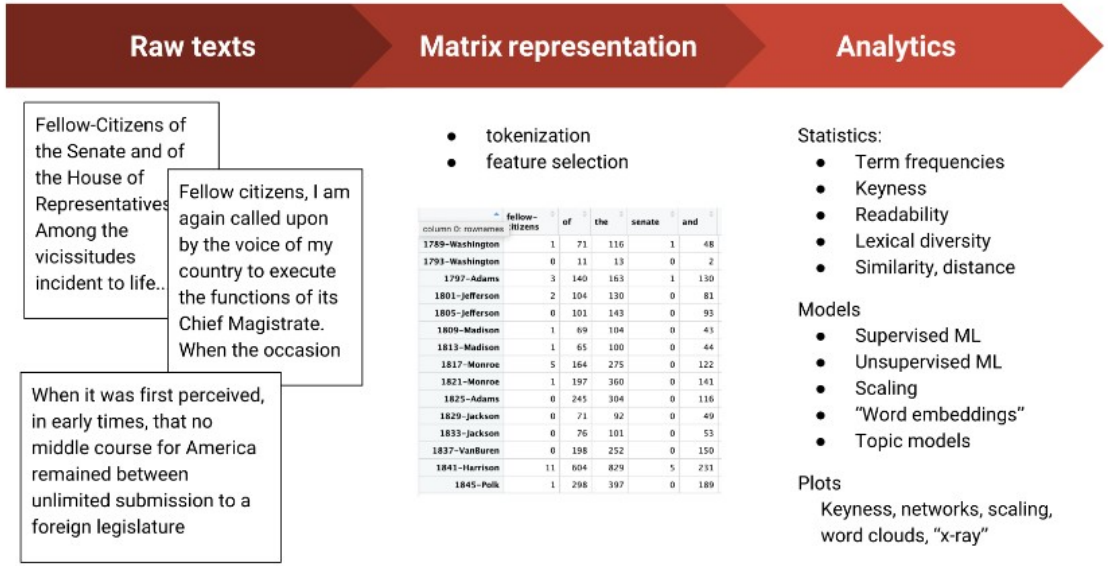
3 Iterate and cumulate

4 Text analysis is low dimensional

5 The best method depends on the task

6 Always validate

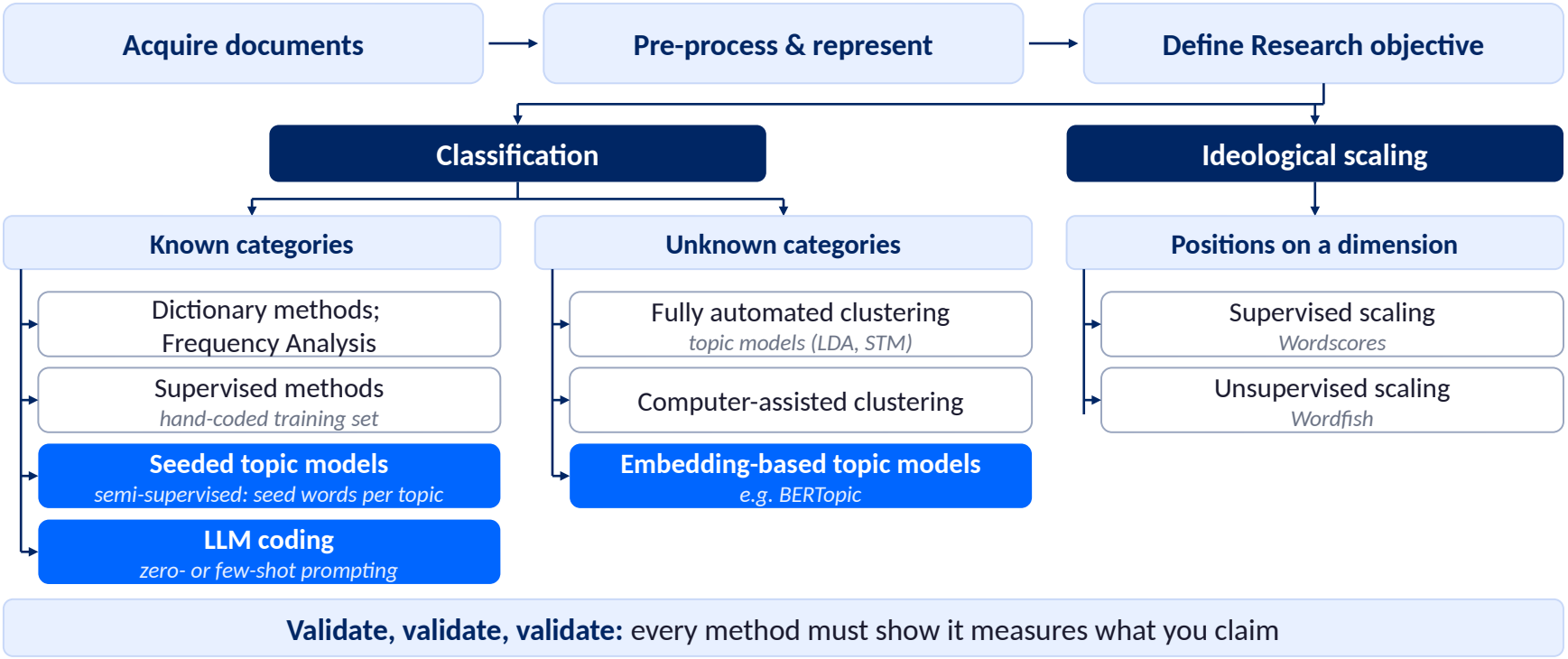
The workflow (simple version), demystified by Kenneth Benoit (Quanteda Initiative)



column 0: docnames	fellow-citizens	of	the	senate	and
1789-Washington	1	71	116	1	48
1793-Washington	0	11	13	0	2
1797-Adams	3	140	163	1	130
1801-Jefferson	2	104	130	0	81
1805-Jefferson	0	101	143	0	93
1809-Madison	1	69	104	0	43
1813-Madison	1	65	100	0	44
1817-Monroe	5	164	275	0	122
1821-Monroe	1	197	360	0	141
1825-Adams	0	245	304	0	116
1829-Jackson	0	71	92	0	49
1833-Jackson	0	76	101	0	53
1837-VanBuren	0	198	252	0	150
1841-Harrison	11	604	829	5	231
1845-Polk	1	298	397	0	189

Source: Benoit 2019; https://www.uni-muenster.de/imperia/md/content/ifpol/grasp/2019-06-27_muenster.pdf

Deciding on a method: A decision-tree



Validate, validate, validate: every method must show it measures what you claim

Adapted and updated from Grimmer & Stewart (2013)

Added by Milan Thies

03

Working with R: Introduction to R & Quanteda

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Did everyone manage to install R, R-Studio and the package quanteda?

A very brief introduction to R

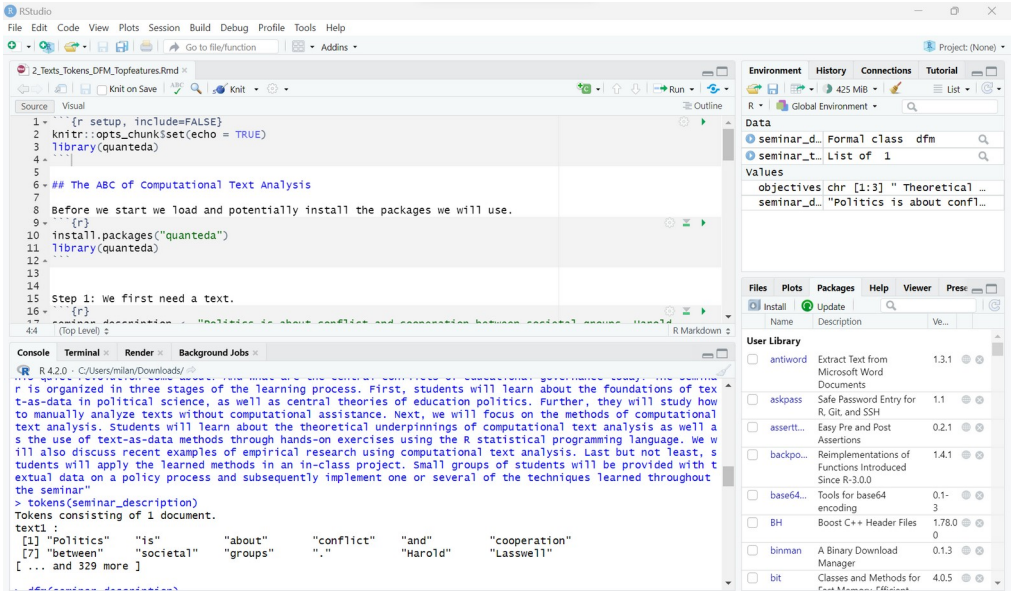
- R is a statistical programming language in which you can perform a large variety of operations
- We will focus on those connected to the analysis of text
- If you do not understand everything while we go through the slides: Do not panic! You will learn it step by step today and I am happy to go through it with you afterwards
- Recommendation: Do not work through the R scripts while I present the steps to you. I know it is tempting but you might miss something important. We will have enough time just for working in the script.

A typical R Session

Environment – objects & values. You can refer to them in your script

Script – tells R what to do. Always save this. Most important window. Here is where you write code.

Console – output. You can find the output of your code here.

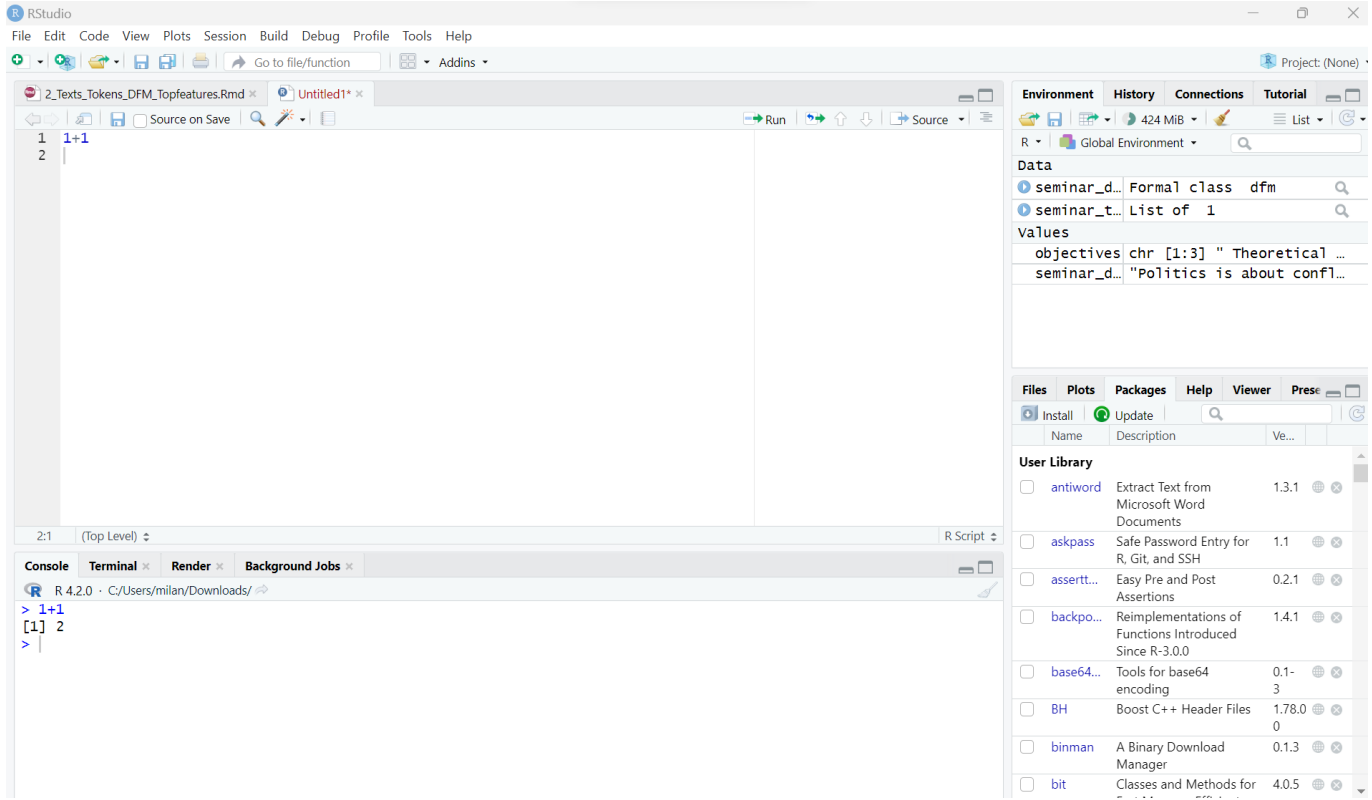


Other information – packages, files etc.

What can we do in Rstudio?

Input – code (= 1+1)

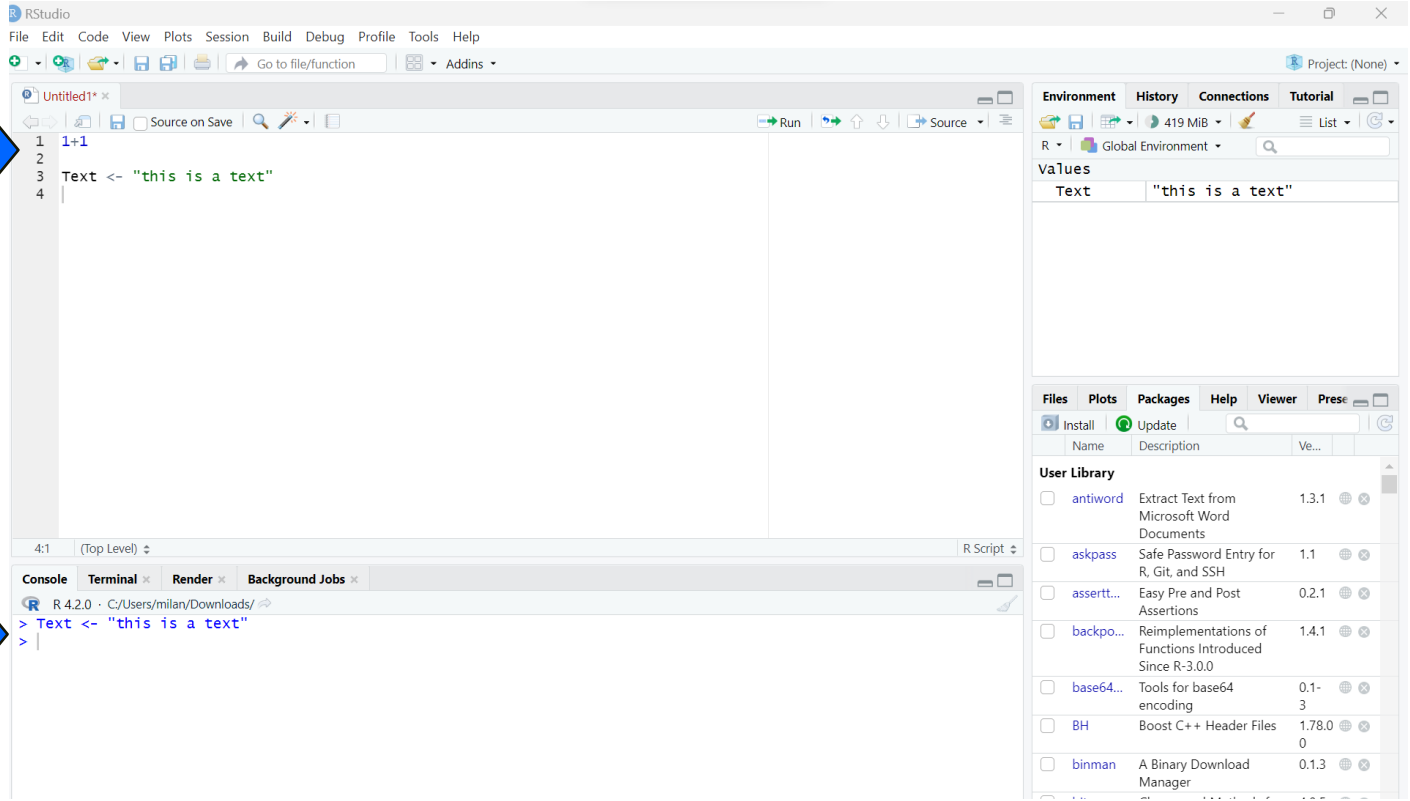
Output – result (= 2)



What can we do in Rstudio?

Input – code (name of an object <- value of the object

Output – operation required by code



What can we do in Rstudio?

- We can do a great variety of things, mostly statistical programming.
- Most importantly for this seminar is that we can analyse text.
- In this seminar we create an object consisting of text documents called **corpus** and then perform operations on the corpus using the package "**quanteda**".
- In R the operations we can run depend on the packages installed and loaded.

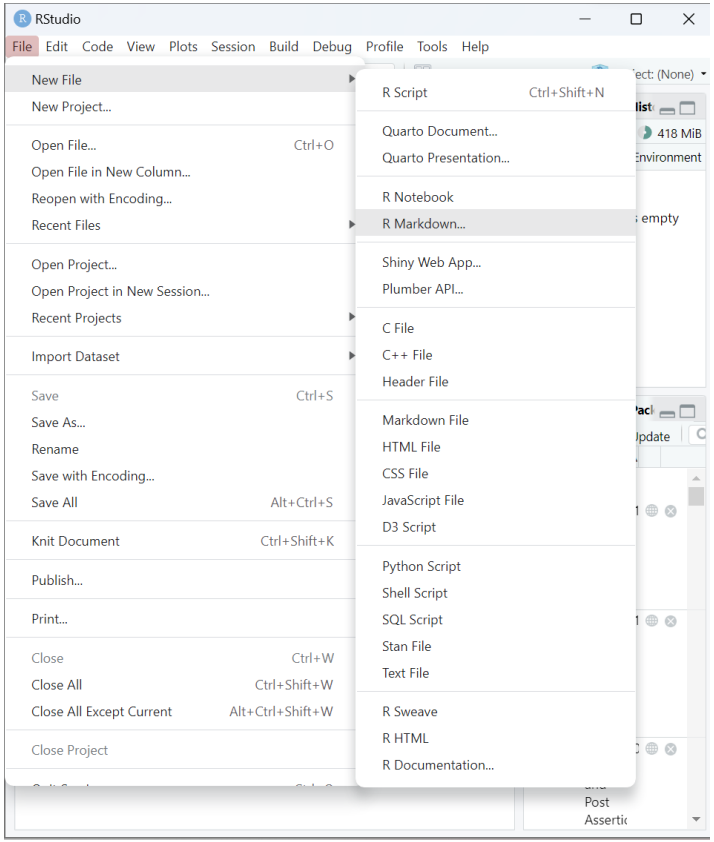


R Markdown

- Instead of writing a normal R script, we will work in R Markdown.
- Markdown is a simple formatting syntax creating HTML, PDF, and MS Word documents including code written for R.
- Why do we work in Markdown? It allows you to see exactly which step of your code produced which output. I highly recommend working in Markdown when you are doing computational text analysis. In the next days you will see why.
    - Disclaimer There are alternatives like Quarto that are equally good
- Further information: http://rmarkdown.rstudio.com

R Markdown

- I will provide Markdown scripts during the seminar, but this is how you create one in RStudio:
  - File
  - New File
  - R Markdown

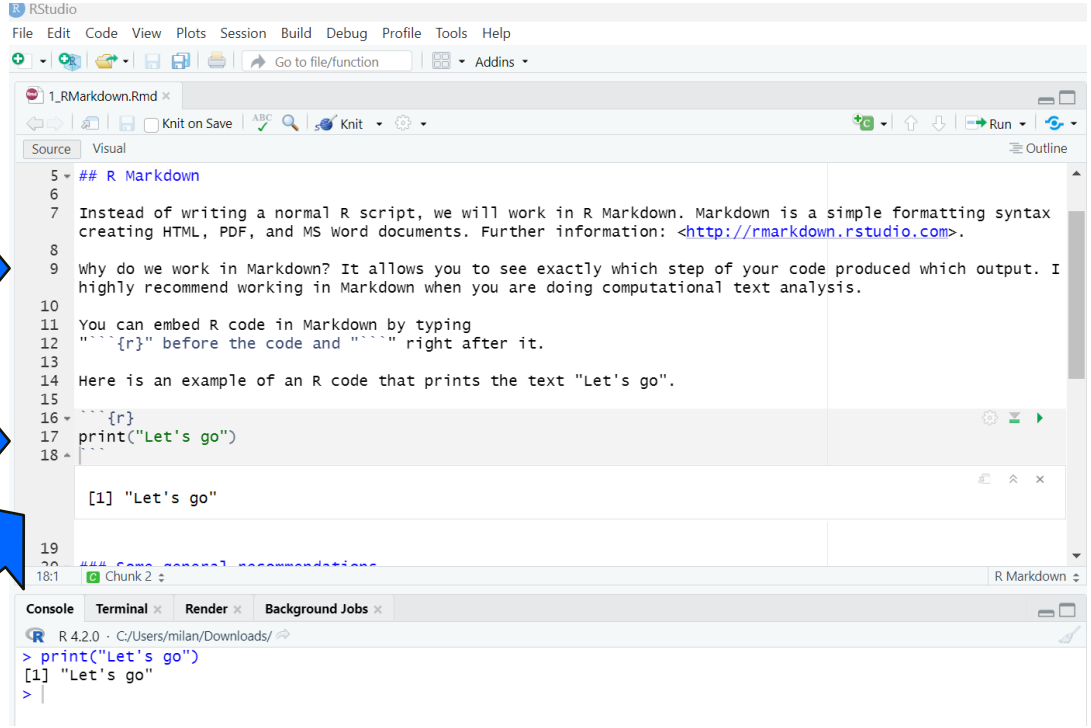


R Markdown

Content – Will appear in the document

Code – Start with ```{r} and end with ```

Output – Will appear in document



All in the script window

Script 1 – R Markdown

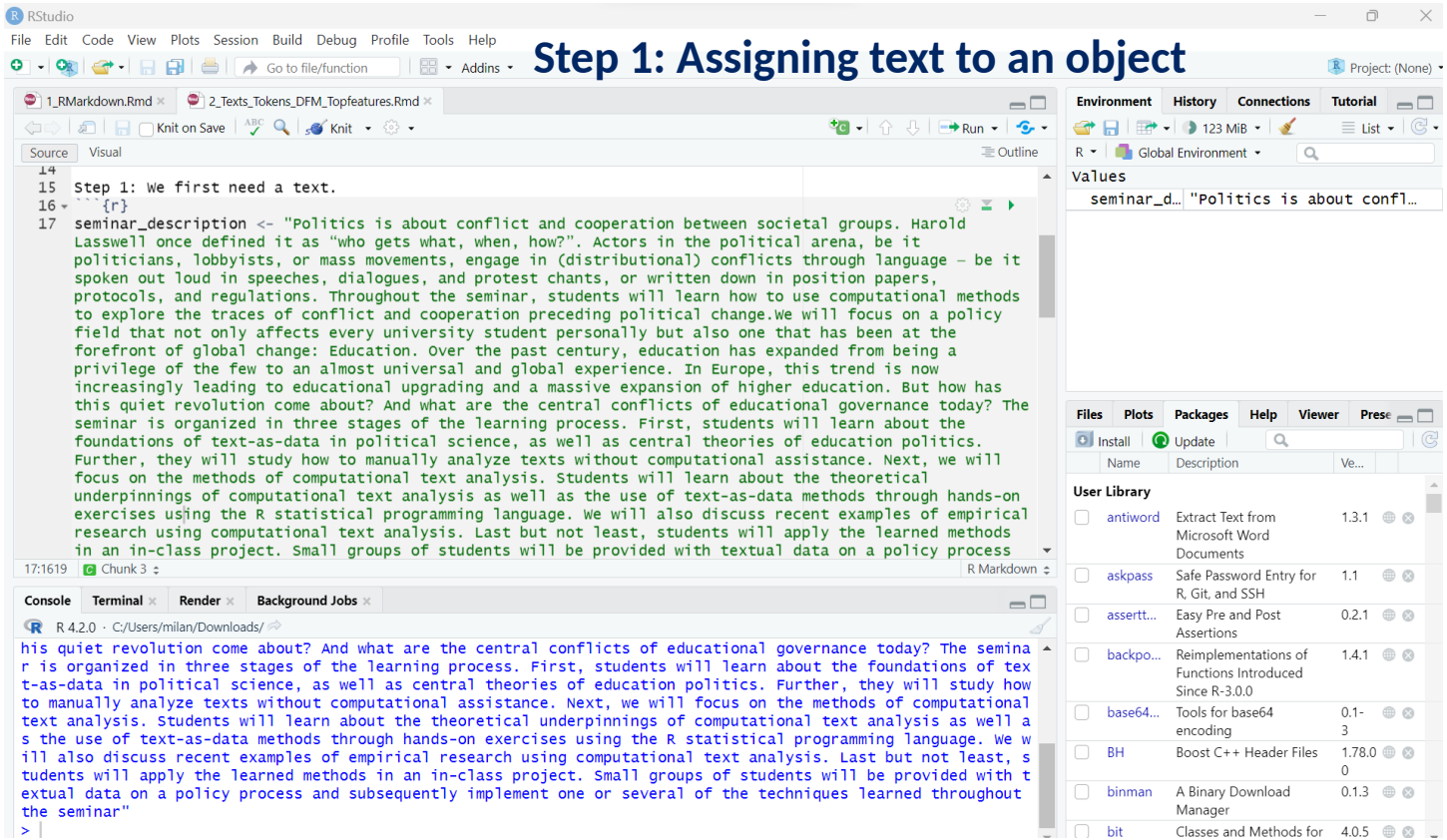
- Open document 1_RMarkdown (find it on my website), understand how the code I wrote for you works.
- If you have any questions, let me know.

The ABC of Computational Text Analysis – Text, Tokens, DFM

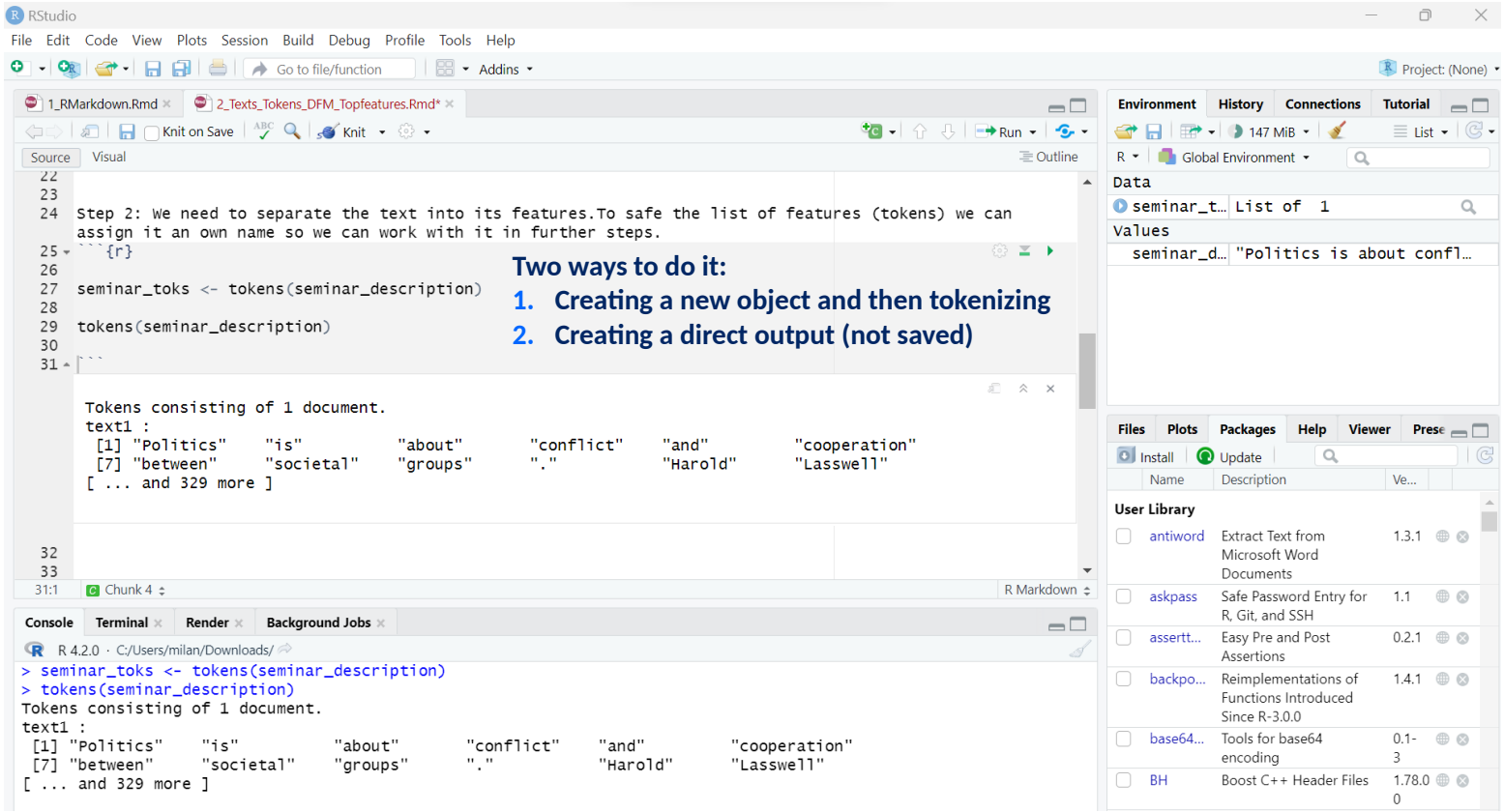
- Who still remembers what tokens are and what a document-feature-matrix is?
- Let’s apply this pipeline in R

The ABC of Computational Text Analysis – Corpus, Tokens, DFM

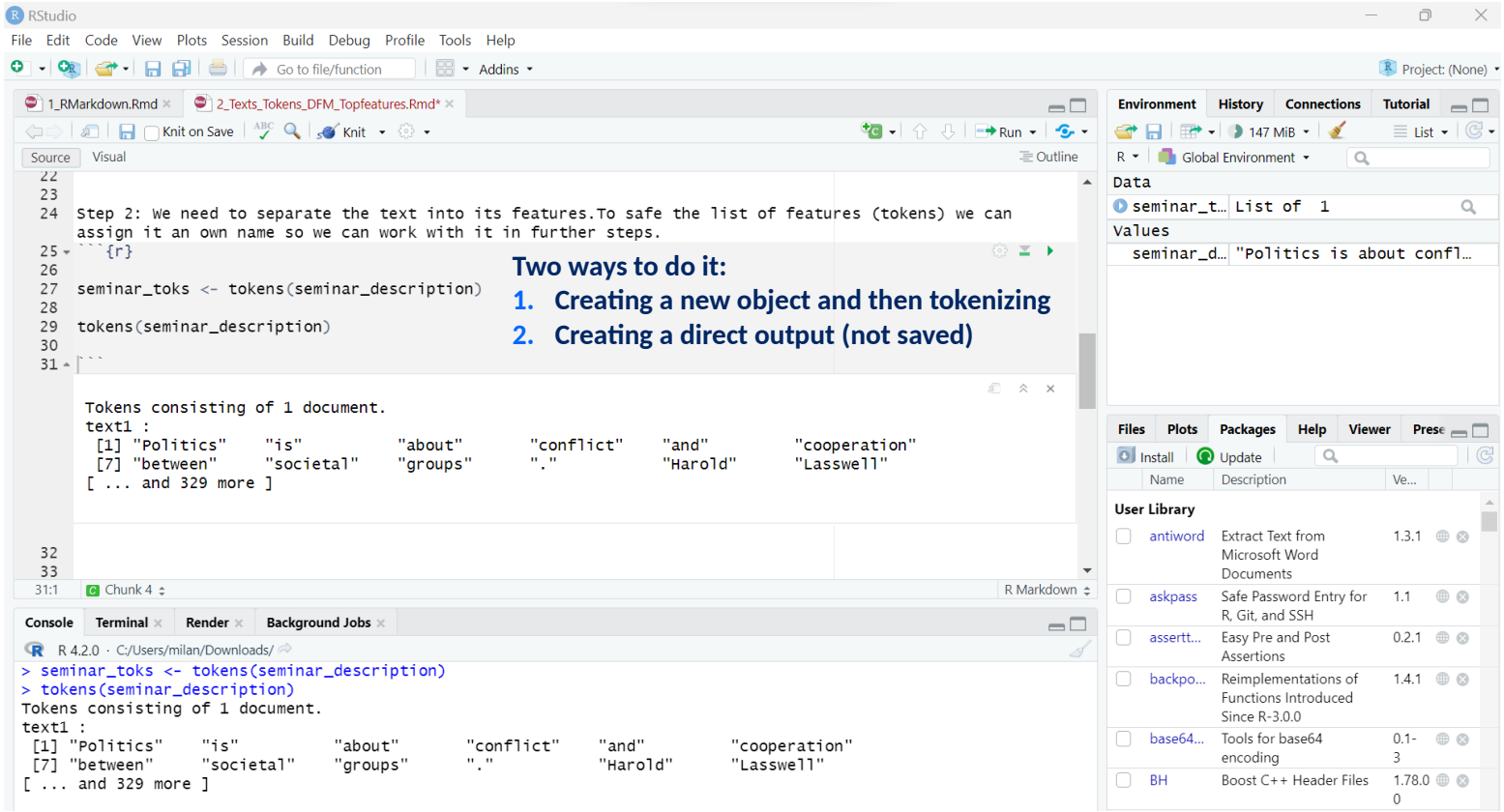
Step 1: Assigning text to an object



Step 2: Tokenization



Step 2: Tokenization



Step 3: Creating a document-feature-matrix

- For many operations you will need a document-feature-matrix (dfm)
- The dfm tells you the frequency per feature for every feature of each document
- Again: 2 ways: creating an object you can work with later (no direct output, option 1) or creating a direct output but no object (option 2)

````
Step 3: We can organise the features in a document-feature-matrix (dfm) which will also give us the frequency of each feature.
```{r}
seminar_dfm <- dfm(seminar_toks)

dfm(seminar_toks)
```
````

```
Document-feature matrix of: 1 document, 176 features (0.00% sparse) and 0 docvars.
       features
docs    politics is about conflict and cooperation between societal groups  .
  text1        2  3     4        2   8           2       1        1      2 13
[ reached max_nfeat ... 166 more features ]
```

Step 4: Topfeatures

- You can also sort the dfm by frequency with the topfeatures function
- Topfeatures are a great first step to understanding what texts are about
- Simply ask for topfeatures of the dfm you created before
- Advanced: You can use %>% (also called pipe) as a way of combining two arguments. Here "create dfm" AND "organise by frequency" (option 2)

````
Step 4: We can learn about the most used features using the **topfeatures** function drawing on the
document-feature-matrix we have created.
```{r}
topfeatures(seminar_dfm)
```

     ,    the     of      .   will    and     in     as students     to
    19     17     14     13      9      8      8      5        5      5

Alternative step 4: We can also combine several steps by using the "%>%" connector. Here we create a dfm
from the tokens we have created (list of features [= mostly words]) and then we also directly ask it to
print the topfeatures (= the most frequent features).
```{r}
dfm(seminar_toks) %>% topfeatures()

```

     ,    the     of      .   will    and     in     as students     to
    19     17     14     13      9      8      8      5        5      5
````

Script 2 – ABC of Text Analysis

- Open document 2_Texts_Tokens_DFM_Topfeatures (find it on my website)
- Work through the script
- If you have any questions, let me know

Almost ready to start analysing text

- Results not really interesting because we find that the most frequent features are not really interesting. What we still need to learn in how to preprocess the data.
- Next section is about your setup and the workflow for text analysis including the preprocessing

04

Workflow & Descriptive Analyses

The real workflow – including pre-processing

- Install & load the packages you will use
- Acquire documents and import them into RStudio
- Pre-Processing
  - Create corpus from dataset
  - Tokenize
  - Create document-feature-matrix
  - Remove non-text (punctuation, symbols, numbers)
  - Remove meaningless words (stopwords, potentially trim)
  - Possibly stem
- Choose method for text analysis
- Plot and visualise

The real workflow – including pre-processing

- Install & load the packages you will use
- Acquire documents and import them into RStudio
- Pre-Processing
  - **Create corpus from dataset: This we did not do in the last script, but we need to do it when we have a dataset with more than just the text.**
  - Tokenize
  - Create document-feature-matrix
  - Learn about data
  - Remove non-text (punctuation, symbols, numbers)
  - Remove meaningless words (stopwords, potentially trim)
  - Possibly stem
- Choose method for text analysis
- Plot and visualise

R

- Who has worked in R before?
- Who knows how to set a working directory?
- Who has installed quanteda before class?

Working Directory

- Before we start: find or set the working directory
- WD is a folder in which RStudio stores output and saves your script
- Download the scrips and data for today from my website and store all in the same folder (= future working directory).
- Then set the working directory in R
- You can set a new working directory with the setwd() function
  - How? You copy the path to a folder you like into the brackets in quotation marks. Make sure to change all \ to /.
  - Example: setwd("C:/Users/…/Data & Scripts")

Website with course materials

- https://milanthies.github.io/teaching/cta-idheap/

Installing and loading packages

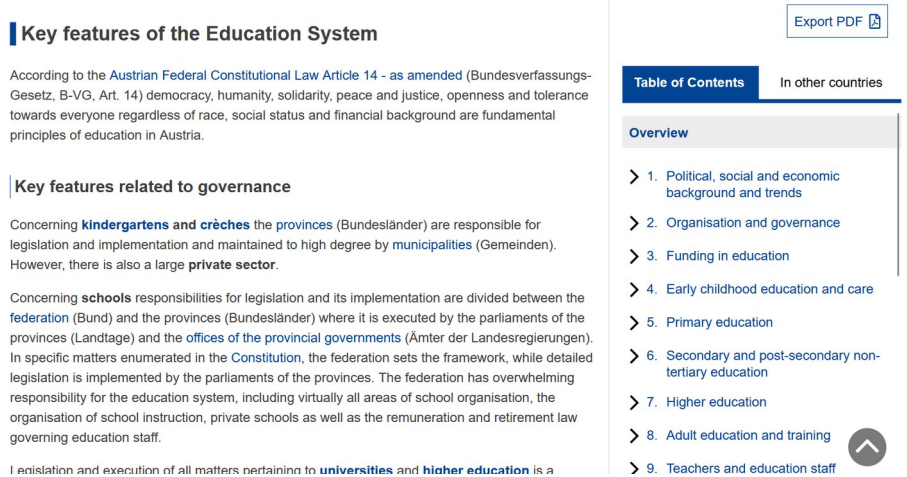
- Next you install the packages you might need and load those you use (through the library function)

```{r}
packages <- c("tidyverse","quanteda","quanteda.textstats",
"quanteda.textplots","quanteda.textmodels","readtext","rmarkdown","knitr")
install.packages(packages)

library(tidyverse)
library(quanteda)
library(quanteda.textmodels)
library(quanteda.textplots)
library(quanteda.textstats)
library(readtext)
```

Acquire documents and import them into RStudio

- Now you can import data into Rstudio
- We now work with Eurydice data
- Governments describe their own education systems at length
- Data set I scraped a few years ago
- No longitudinal data, but great standardized cross-sectional data
- Nicely organised into table for you to work with



Key features of the Education System

According to the Austrian Federal Constitutional Law Article 14 - as amended (Bundesverfassungs-Gesetz, B-VG, Art. 14) democracy, humanity, solidarity, peace and justice, openness and tolerance towards everyone regardless of race, social status and financial background are fundamental principles of education in Austria.

Key features related to governance

Concerning **kindergartens and crèches** the provinces (Bundesländer) are responsible for legislation and implementation and maintained to high degree by municipalities (Gemeinden). However, there is also a large **private sector**.

Concerning **schools** responsibilities for legislation and its implementation are divided between the federation (Bund) and the provinces (Bundesländer) where it is executed by the parliaments of the provinces (Landtage) and the offices of the provincial governments (Ämter der Landesregierungen). In specific matters enumerated in the Constitution, the federation sets the framework, while detailed legislation is implemented by the parliaments of the provinces. The federation has overwhelming responsibility for the education system, including virtually all areas of school organisation, the organisation of school instruction, private schools as well as the remuneration and retirement law governing education staff.

Legislation and execution of all matters pertaining to **universities** and **higher education** is a

Export PDF

Table of Contents | In other countries

Overview

1. Political, social and economic background and trends
2. Organisation and governance
3. Funding in education
4. Early childhood education and care
5. Primary education
6. Secondary and post-secondary non-tertiary education
7. Higher education
8. Adult education and training
9. Teachers and education staff

Acquire documents and import them into RStudio: Eurydice

- Data set available for download on my website
- Find out the path to the document. Does everybody know how to find the path?
- Steps:
  - Copy the path to the RDS document "eurydice“.
  - Use the readRDS() function to import it into the environment.
  - Create a new object (systems)
  - Create a corpus that only includes the features of the RDS imported.

```r
systems <- readRDS("C:/Users/.../eurydice.RDS")


systems_corp <- corpus(systems)
```

Eurydice in R

	country	chapter	text	url
1	Albania	Political,Social and Economic Background and Trends	This chapter discusses political, social and economic backgr...	https://eacea.ec.europa
2	Albania	Historical Development	Albanian territories were inhabited since 100,000 years ago....	https://eacea.ec.europa
3	Albania	Main Executive and Legislative Bodies	The President and the Assembly The President of the Repub...	https://eacea.ec.europa
4	Albania	Population: Demographic Situation, Languages and Religion...	Population and Demographic Situation The Republic of Alba...	https://eacea.ec.europa
5	Albania	Political and Economic Situation	Albanian is classified as a middle-income country since 200...	https://eacea.ec.europa
6	Albania	Organisation and Governance	This chapter is divided into six sub-sections the first discuss...	https://eacea.ec.europa
7	Albania	Fundamental Principles and National Policies	Fundamental principles of national education system in the ...	https://eacea.ec.europa
8	Albania	Lifelong learning strategy	Lifelong learning has been a cross-cutting issue, addressed ...	https://eacea.ec.europa
9	Albania	Organisation of the education system and of its structure	Pre-university education system. Organization. Based on law...	https://eacea.ec.europa
10	Albania	Organisation of private education	Private pre-university education Likewise with public Pre-uni...	https://eacea.ec.europa
11	Albania	National qualifications framework	Albanian Qualifications Framework was adopted by Law No....	https://eacea.ec.europa
12	Albania	Administration and governance at central and/or regional le...	Pre university education According to the pre-university edu...	https://eacea.ec.europa
13	Albania	Administration and governance at local and/or institutional ...	Pre- University education The Law on pre-university educati...	https://eacea.ec.europa

Pre-Processing: Tokenize

- At word, sentence or character level. We will tokenize at word level and store tokenized text in new object.
- Why not at character level?

```
50 ```{r}
51 tokens(systems_corp,"word")
52 tokens(systems_corp ,"sentence")
53 tokens(systems_corp,"character")
54 systems_toks <- tokens(systems_corp)
55
56 ```
```

46:16 # Tokenization R Markdown

Console Terminal Background Jobs

R 4.2.0 · ~/

```
Tokens consisting of 4,465 documents and 3 docvars.
text1 :
 [1] "This"       "chapter"    "discusses"  "political"  ","          "social"     "and"
 [8] "economic"   "background" "and"        "trends"     "of"
[ ... and 59 more ]

text2 :
 [1] "Albanian"    "territories" "were"        "inhabited"   "since"       "100,000"
 [7] "years"       "ago"         "."           "Over"        "the"         "centuries"
[ ... and 409 more ]

text3 :
```

Pre-Processing: Document-Feature-Matrix

- You know the drill: Use the dfm() function and store in new object

```{r}
systems_dfm <- dfm(systems_toks)
systems_dfm
```

```
Document-feature matrix of: 4,465 documents, 71,392 features (99.49% sparse) and 3 docvars.
       features
docs    this chapter discusses political  , social and economic background trends
  text1    1       2         1         2  1      1   5        2          1      1
  text2    1       0         0         1 20      4  17        3          0      0
  text3    0       0         0         0  4      0  22        0          0      0
  text4    2       0         0         0 34      0  14        0          0      0
  text5    0       0         0         0  5      0   5        2          0      0
  text6    1       2         3         0  0      0   2        0          0      0
[ reached max_ndoc ... 4,459 more documents, reached max_nfeat ... 71,382 more features ]
```

Get to know the data: Keywords in context

- Get window of word before and after a keyword
- How?

```{r}
kwic(systems_toks, "education",window=4)
```

- What do we see?

<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
[text3, 354]	. The Ministry of	|	Education	|	, Sports and Youth
[text3, 375]	in charge for drafting	|	education	|	policies in the country
[text3, 386]	the area of pre-univeristy	|	education	|	it has full competences
[text3, 397]	the area of higher	|	education	|	it shares competences with
[text3, 403]	shares competences with higher	|	education	|	institutions in compliance with
[text5, 140]	%. Expenses for	|	education	|	in 2019 as percentage
[text5, 164]	INSTAT. Expenses for	|	education	|	in 2020 account to
[text6, 20]	policies that rule the	|	education	|	system in Albania.

Get to know the data:

- We can analyze the number of tokens (=words) per document
- The number of features or documents in the dfm
- The most frequent features before starting to clean the data

1. Number of tokens per document

```
```{r}
ntoken(systems_toks) %>% head()
```
```

```
text1 text2 text3 text4 text5 text6
   71   421   410   455   175    79
```

2. The number of feature in our dfm

```
```{r}
nfeat(systems_dfm)
```
```

```
[1] 60998
```

3 Steps of Pre-Processing:

1. Remove non-text
2. Remove uninformative text
3. Unite features (depending on method)

- Removing non-text means removing punctuation, numbers and symbols
- In process of tokenization, we can ask to remove each feature: remove_punct = T means that it is true (T) that we want to remove punctuation.
- We need to create dfm from clean tokens again

```
121  systems_toks <- tokens(systems_corp,  remove_punct=T,
122    remove_numbers=T, remove_symbols=T)
123  systems_dfm <- dfm(systems_toks)
```

3 Steps of Pre-Processing: Step 2: Remove uninformative text

- Texts often contain words that are not interesting or even harmful for our analysis. See for example the topfeatures before pre-processing:

the	of	and	in	to	education	for	a	is	are
408922	251546	218485	138468	117839	105486	99107	86702	58311	47690

- These words are with one exception stopwords. Let's remove them!

```
systems_dfm <- dfm_remove(systems_dfm,stopwords("en"))
```

- There are lists of stopwords for many languages, include all that matter.
- We can also remove words using the dfm_trim() function. Example: words that appear too often to be relevant or words that appear too few times to matter.

3 Steps of Pre-Processing: Step 3: Unite features

- If we reduce the features to their wordstem, we unite words/features with the same stem, often sharing the same meaning. E.g. "systems" and "system" or "programm" and "programme"

```
systems_dfm <- dfm_wordstem(systems_dfm,"en")
```

Now you: Script 3

- Work on Script “3_Pre-Processing & Descriptives”
- **Work at your own pace, ask me or your peers for help**
- Do not start with the “Advanced Transformations” and the “Descriptive Analysis” section yet

How did it go? How far did you get?

Use existing document variables

- Our initial data set had variables that describe the text: Information about the text. Which country does the text describe? Which part of the education system does the text describe?
- We want to use that information in the analysis. corpus() keeps it as document variables (docvars).
- Let's check and use the docvars

	country	chapter	text	url
1	Albania	Political,Social and Economic Background and Trends	This chapter discusses political, social and economic backgr...	https://eacea.ec.europa
2	Albania	Historical Development	Albanian territories were inhabited since 100,000 years ago....	https://eacea.ec.europa
3	Albania	Main Executive and Legislative Bodies	The President and the Assembly The President of the Repub...	https://eacea.ec.europa
4	Albania	Population: Demographic Situation, Languages and Religion...	Population and Demographic Situation The Republic of Alba...	https://eacea.ec.europa
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8	Albania	Lifelong learning strategy	Lifelong learning has been a cross-cutting issue, addressed ...	https://eacea.ec.europa
9	Albania	Organisation of the education system and of its structure	Pre-university education system. Organization. Based on law...	https://eacea.ec.europa
10	Albania	Organisation of private education	Private pre-university education Likewise with public Pre-uni...	https://eacea.ec.europa
11	Albania	National qualifications framework	Albanian Qualifications Framework was adopted by Law No....	https://eacea.ec.europa
12	Albania	Administration and governance at central and/or regional le...	Pre university education According to the pre-university edu...	https://eacea.ec.europa
13	Albania	Administration and governance at local and/or institutional ...	Pre- University education The Law on pre-university educati...	https://eacea.ec.europa

```
docvars(systems_dfm, "country") <- systems$country
docvars(systems_dfm, "chapter") <- systems$chapter
```

```
systems_… Large dfm (220995175…
    ..@ docvars :'data.frame':…
    .. ..$ docname_: chr [1:4465]…
    .. ..$ docid_  : Factor w/ 44…
    .. ..$ segid_  : int [1:4465]…
    .. ..$ country : chr [1:4465]…
    .. ..$ chapter : chr [1:4465]…
```

Advanced Transformations: Group documents

- After attaching the country and chapter variable we can analyze text per country or chapter by using the dfm_group function
- This helps if we want to analyze the education system of one country or compare one sector of the education system across countries

```
```{r}
country_system_dfm <- dfm_group(systems_dfm,country)
chapter_system_dfm <- dfm_group(systems_dfm,chapter)

```
```

```
country_system_d… Large dfm (2128285 elements,  5.8 MB)
    ..@ docvars :'data.frame':    43 obs. of  4 variables:
    .. ..$ docname_: chr [1:43] "Albania" "Austria" "Belgium - Fle
    .. ..$ docid_  : Factor w/ 43 levels "Albania","Austria",..: 1
    .. ..$ segid_  : int [1:43] 1 1 1 1 1 1 1 1 1 1 ...
    .. ..$ country : chr [1:43] "Albania" "Austria" "Belgium - Fle
```

Advanced Transformations: Create a subset

- Sometimes we want to analyse only some part of the data
- Therefore, we need to create a subset of the data
- Let's create a subset in which we exclude Germany and one in which we only keep data on Germany with dfm_subset()

```
nogermany_system_dfm <- dfm_subset(systems_dfm,country!="Germany")
Germany_dfm <- dfm_subset(systems_dfm, country == "Germany")
```

Descriptive Analyses

What we will cover

- How to conduct and visualise simple text-as-data analyses
- Analyses based on absolute word frequencies: Topfeatures and Wordclouds
- Analysis based on relative word frequencies (keyness / keyword analysis)
  - How frequent are words compared to the overall corpus?
- What we will cover later in the seminar:
  - Frequencies of selected words
  - Dictionaries
  - Topic Models
  - In-depth discussion on best practices and reading (Martin, 2018)

Topfeatures

- We have seen the mechanics of how to get topfeatures earlier today. Let's compare how useful this is to understand what a text is about before and after pre-processing.
- Before:

the	of	and	in	to	education	for	a	is	are
408922	251546	218485	138468	117839	105486	99107	86702	58311	47690

- After:

educ	school	student	institut	train	programm
125054	71368	27918	27828	25322	23814

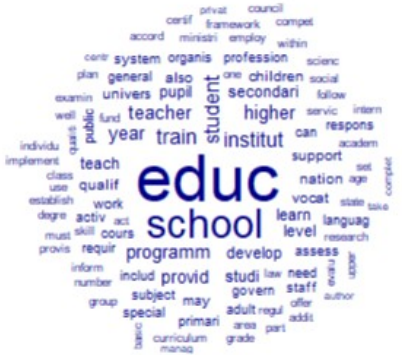
teacher	higher	year	provid
23326	23166	22670	19144

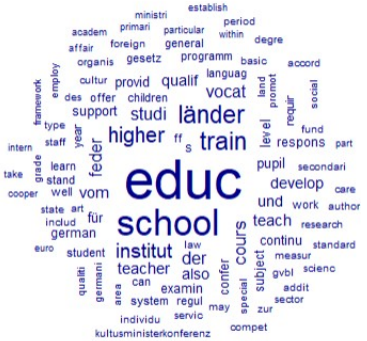
Wordclouds

- Wordclouds are perhaps the most popular visualization of absolute frequencies
- We can create them with a simple line of code:

```
textplot_wordcloud(systems_dfm, max_words = 100)
```

- We can define the number of words the cloud displays, and we can also subset our dfm to create and compare wordclouds for different texts
- What differences do you see between the overall cloud and the one for Germany?





Keyword analysis / keyness

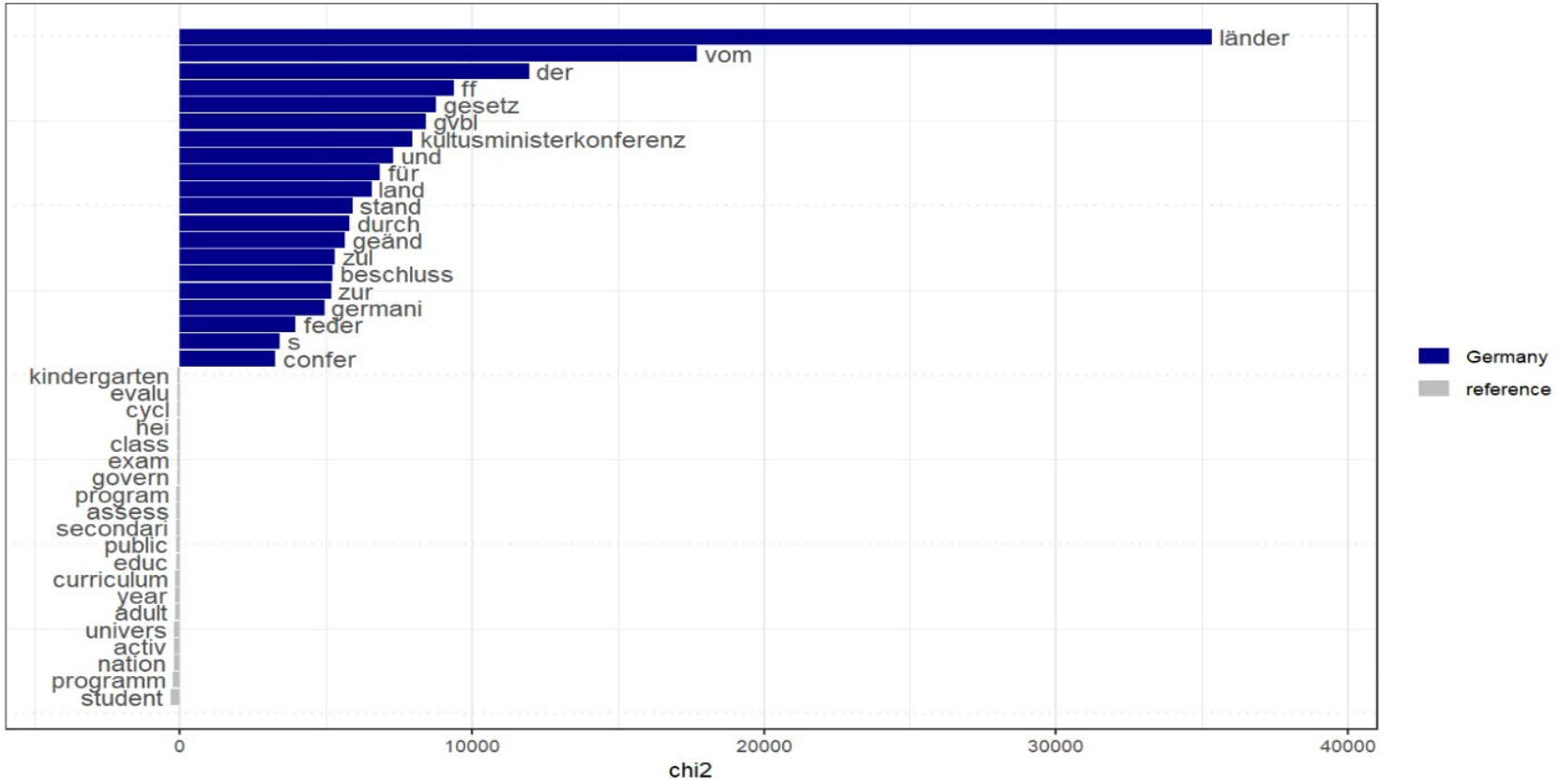
- Relative word frequency: Words that appear more frequent in one text / a group of texts compared to the overall corpus
- Great for understanding differences
- However, easy to confound by particular words (e.g. in another language).
- How to?
  - Define how to group documents
  - Pipe into textstat_keyness()
  - Define group of interest
  - head() gives you a list of 10 top words
  - textplot_keyness() gives you a graph

```
keyness <- dfm_group(systems_dfm,country) %>%
 textstat_keyness("Germany")

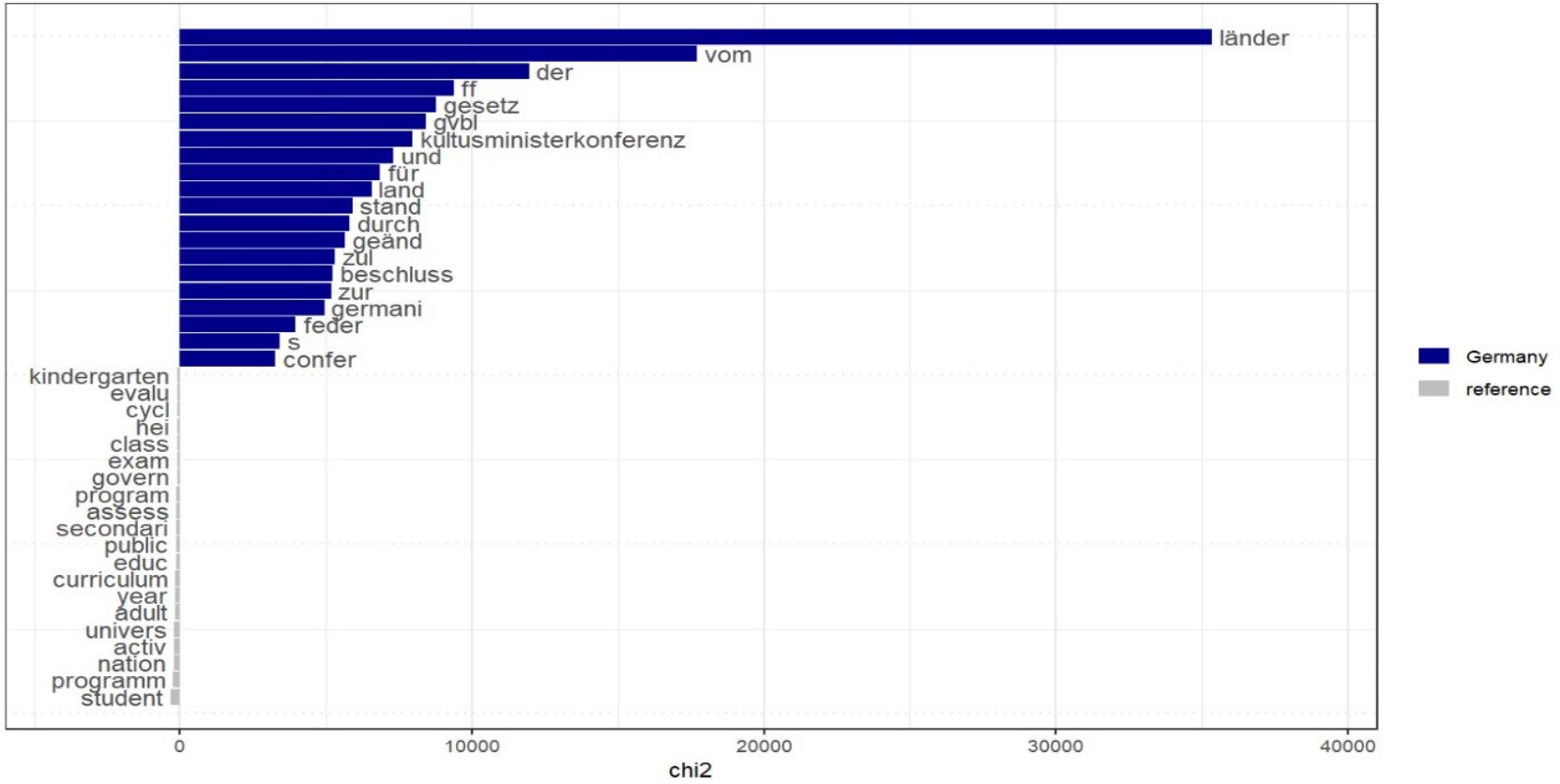
head(keyness)

keyness %>%
 textplot_keyness()
```

How can we interpret this?



How can we interpret this?



Now you: Script 3 - continued

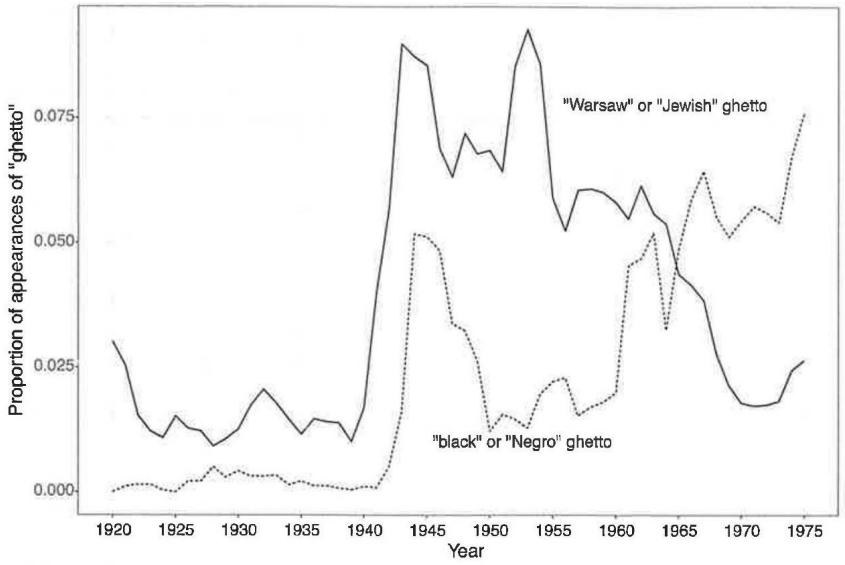
- Work on Script “3_Pre-Processing & Descriptives”
- Work at your own pace, **ask me or your peers for help**
- In the end you will be able to work with other data, pre-processing it, running analyses and visualizing it

05

Word Counts and Dictionaries

Word Counting

- The simplest form of the dictionary approach: count how often a word or phrase appears
- Advantage: clear and easy to communicate
- Problem: complex concepts are hard to capture with single words
- With carefully chosen words and validation by careful reading, word counts can be extraordinarily effective
- Example: Duneier (2016) traces two meanings of the word “ghetto” with Google Ngrams (appearing in books, newspapers etc)



Duneier (2016), Graph 1, in Grimmer, Roberts & Stewart (2022), Fig. 16.1

What are dictionaries?

- **Keys** that stand for a theorized concept
- **Values** that represent the keys empirically

- Example: Food (key); bread, apple, banana, chocolate, potatoes (values)

- How to measure the prevalence of a concept in a corpus?
  - We count the frequency of dictionary features (values)

Using dictionaries

- **Advantages:**
  - Easy to learn
  - Time and cost-efficient
- **Disadvantages:**
  - Human involvement might cause bias
  - Creating new dictionaries can be difficult and lead to biased findings
  - Dictionary analyses depend on individual words. In small data sets individual words can have large effects. Also: Problem of negation: “eat animals” or “don’t eat animals”
- **Challenge**: Dictionaries must be exhaustive but also unambiguous. Dictionary creation must be transparent

Existing Dictionaries

- Ideologies - Pauwels (2011):
  Measuring Populism: A Quantitative Text Analysis of Party Literature in Belgium
- Uses frequency of word use to measure if text expresses ideology

Table A2. Dictionary

Dictionary	Dutch words	Translation
Conservatism	christ*; geloof; gezin; kerk; normen; porn*; seks*; waarden	christ*; belief; family; church; norm; porn*; sex*; values
Environment	ecol*; groene*; klimaat*; milieu*; opwarming	ecol*; green*; climate*; environment*; heating
Immigration	marok*; turk; allocht*; asiel*; halal*; hoofddoek*; illega*; immigr*; islam*; koran; moslim*; vreemd*	moroc*; turk; allocht*; asylum*; halal*; scarf*; illega*; immigr*; islam*; koran; muslim*; foreign*

Existing Dictionaries

- Are recommendation letters gendered?
- Schmader et al. (2007): A Linguistic Comparison of Letters of Recommendation for Male and Female Chemistry and Biochemistry Job Applicants

Study-Defined Dimension Dictionaries Standout words: excellen*, superb, outstanding, unique, exceptional, unparalleled, *est, most, wonderful, terrific*, fabulous, magnificent, remarkable, estraordinar*, amazing, supreme*, unmatched

Ability words: talent*, intell*, smart*, skill*, ability, genius, brilliant*, bright*, brain*, aptitude, gift*, capacity, propensity, innate, flair, knack, clever*, expert*, proficient*, capable, adept*, able, competent, natural*, inherent*, instinct*, adroit*, creative*, insight*, analytical

Grindstone words: hardworking, conscientious, depend*, meticulous, thorough, diligen*, dedicate, careful, reliab*, effort*, assiduous, trust*, responsib*, methodical, industrious, busy, work*, persist*, organiz*, disciplined

Teaching words: teach, instruct, educat*, train*, mentor, supervis*, adviser, counselor, syllabus, syllabus, course*, class, service, colleague, citizen, communicate*, lectur*, student*, present*, rapport

Research words: research*, data, study, studies, experiment*, scholarship, test*, result*, finding*, publication*, publish*, vita*, method*, scien*, grant*, fund*, manuscript*, project*, journal*, theor*, discover*, contribution*

Disclaimer:

Too much code and too quick? Do not worry, you go through the same code with lots of time

Workflow: Working with existing dictionaries – Step 1: Load data and Pre-Process

1.
```
debate_data <- read.csv("C:/Users/.../us_election_2020_1st_presidential_debate.csv")
```
2.
```
corp_debate <- corpus(debate_data)
```
3.
```
toks_debate <- tokens(corp_debate,  remove_punct=T,  remove_numbers=T, remove_symbols=T)
```
4.
```
dfm_debate <- dfm(toks_debate)
```
5. & 6.
```
dfm_debate <- dfm_remove(dfm_debate,stopwords("en"))
dfm_debate <- dfm_wordstem(dfm_debate,"en")
```

What do we do when running the code lines 1-6?

Data: US Presidential Election Debate 2020

	speaker	minute	text
1	Chris Wallace	01:20	Good evening from the Health Education Campus of Case Western Reserve University and the Cleveland Clinic. I'm Chris Wallace of Fo
2	Chris Wallace	02:10	This debate is being conducted under health and safety protocols designed by the Cleveland Clinic, which is serving as the Health Secu
3	Vice President Joe Biden	02:49	How you doing, man?
4	President Donald J. Trump	02:51	How are you doing?
5	Vice President Joe Biden	02:51	I'm well.
6	Chris Wallace	03:11	Gentlemen, a lot of people been waiting for this night, so let's get going. Our first subject is the Supreme Court. President Trump, you r
7	President Donald J. Trump	04:01	Thank you very much, Chris. I will tell you very simply. We won the election. Elections have consequences. We have the Senate, we have
8	President Donald J. Trump	04:53	And we won the election and therefore we have the right to choose her, and very few people knowingly would say otherwise. And by t
9	Chris Wallace	05:22	President Trump, thank you. Same question to you, Vice President Biden. You have two minutes.
10	Vice President Joe Biden	05:29	Well, first of all, thank you for doing this and looking forward to this, Mr. President.
11	President Donald J. Trump	05:34	Thank you, Joe.
12	Vice President Joe Biden	05:36	The American people have a right to have a say in who the Supreme Court nominee is and that say occurs when they vote for United S
13	Vice President Joe Biden	06:12	Now, what's at stake here is the President's made it clear, he wants to get rid of the Affordable Care Act. He's been running on that, he
14	Vice President Joe Biden	07:08	And that ended when we, in fact, passed the Affordable Care Act, and there's a hundred million people who have pre-existing conditio
15	President Donald J. Trump	07:34	There aren't a hundred million people with pre-existing conditions. As far as a say is concerned, the people already had their say. Okay,
16	Vice President Joe Biden	08:01	He's elected to the next election.
17	President Donald J. Trump	08:02	During that period of time, during that period of time, we have an opening. I'm not elected for three years. I'm elected for four years. J

Workflow: Dictionaries – Step 2: Load Dictionary

- If working with pre-existing dictionary, save dictionary in working directory
- Load with dictionary function
- Here: Newsmap geography dictionary

```
newsmap_dict <- dictionary(file = "english.yml",
                           format = "YAML")
```

```
newsmap_dict
```

```
## Dictionary object with 5 primary key entries and 3 nested levels.
## - [AFRICA]:
##   - [EAST]:
##     - [BI]:
##       - burundi, burundian*, bujumbura
##     - [DJ]:
##       - djibouti, djiboutian*
##     - [ER]:
##       - eritrea, eritrean*, asmara
##     - [ET]:
##       - ethiopia, ethiopian*, addis ababa
##     - [KE]:
##       - kenya, kenyan*, nairobi
##     - [KM]:
##       - comoros, comorian*, moroni
##     [ reached max_nkey ... 13 more keys ]
##   - [MIDDLE]:
##     - [AO]:
##       - angola, angolan*, luanda
##     - [CD]:
##       - democratic republic congo, dr congo, drc, democratic republic congolese, dr congolese, kinshasa
```

Workflow: Dictionaries – Step 2: Load Dictionary

- Words in dictionary: countries, nationalities, capitals
- What kind of questions could we answer with this dictionary?

```
newsmap_dict
```

```
## Dictionary object with 5 primary key entries and 3 nested levels.
## - [AFRICA]:
##   - [EAST]:
##     - [BI]:
##       - burundi, burundian*, bujumbura
##     - [DJ]:
##       - djibouti, djiboutian*
##     - [ER]:
##       - eritrea, eritrean*, asmara
##     - [ET]:
##       - ethiopia, ethiopian*, addis ababa
##     - [KE]:
##       - kenya, kenyan*, nairobi
##     - [KM]:
##       - comoros, comorian*, moroni
##     [ reached max_nkey ... 13 more keys ]
##   - [MIDDLE]:
##     - [AO]:
##       - angola, angolan*, luanda
##     - [CD]:
##       - democratic republic congo, dr congo, drc, democratic republic congolese, dr congolese, kinshasa
```

Workflow: Dictionaries – Step 3: Apply Dictionary

- Use dfm_lookup() command to match dictionary words with dfm
- What do we see here? For each document we test if a feature from the dictionary is present or not

```
dict_dfm_results <- dfm_lookup(dfm_debate,newsmap_dict)
dict_dfm_results[610:615,111:119]
```

```
## Document-feature matrix of: 6 documents, 9 features (96.30% sparse) and 2 docvars.
##          features
## docs      AMERICA.NORTH.GL AMERICA.NORTH.PM AMERICA.NORTH.US ASIA.CENTER.KG
##   text610                0                0                2              0
##   text611                0                0                0              0
##   text612                0                0                1              0
##   text613                0                0                0              0
##   text614                0                0                0              0
##   text615                0                0                0              0
##          features
## docs      ASIA.CENTER.KZ ASIA.CENTER.TJ ASIA.CENTER.TM ASIA.CENTER.UZ
##   text610              0              0              0              0
##   text611              0              0              0              0
##   text612              0              0              0              0
##   text613              0              0              0              0
##   text614              0              0              0              0
##   text615              0              0              0              0
##          features
```

Workflow: Dictionaries – Step 4: Frequency analysis

- Using the textstat_frequency() command we can get the frequency and a ranking of the prevalence of regions
- So: What might be the geopolitical focus of the 2020 US presidential campaigns?
- We can apply dictionaries to either dfms or tokens

```
dict_dfm_results %>% textstat_frequency()
```

```
##              feature frequency rank docfreq group
## 1   AMERICA.NORTH.US        44    1      35   all
## 2       ASIA.EAST.CN        10    2       9   all
## 3     EUROPE.EAST.RU         6    3       6   all
## 4      ASIA.SOUTH.IN         2    4       2   all
## 5  AMERICA.CENTER.MX         1    5       1   all
## 6   AMERICA.SOUTH.BR         1    5       1   all
## 7       ASIA.EAST.JP         1    5       1   all
## 8       ASIA.WEST.IQ         1    5       1   all
## 9    EUROPE.NORTH.IE         1    5       1   all
## 10    EUROPE.WEST.DE         1    5       1   all
```

Beyond frequencies: Dictionaries with levels

- Some dictionaries have levels
- Newsmap has 3 levels: Continents, regions and countries
- In script, you can explore levels further

```
dfm_debate %>% dfm_lookup(newsmap_dict, levels = 2) %>%
  textstat_frequency()
```

```
##   feature frequency rank docfreq group
## 1   NORTH        45    1      35   all
## 2    EAST        17    2      13   all
## 3   SOUTH         3    3       3   all
## 4    WEST         2    4       2   all
## 5  CENTER         1    5       1   all
```

Analysing by group: Comparing Trump's and Biden's geopolitical focus

```
dfm_debate %>% dfm_lookup(newsmap_dict, levels = 1) %>%
  textstat_frequency(groups=speaker)
```

```
##   feature frequency rank docfreq                    group
## 1 AMERICA        11    1      10             Chris Wallace
## 2    ASIA        11    1       8 President Donald J. Trump
## 3 AMERICA         8    2       6 President Donald J. Trump
## 4  EUROPE         7    3       7 President Donald J. Trump
## 5 AMERICA        27    1      20  Vice President Joe Biden
## 6    ASIA         3    2       3  Vice President Joe Biden
## 7  EUROPE         1    3       1  Vice President Joe Biden
```

And now you!

- Download script “4_Dictionaries”, the file “us_election_2020_1st_presidential_debate” and the dictionary “english.yml”
- We have covered everything up to “dictionary creation”

Dictionary creation

- Technically (like below): dictionary(list(a = c("x"), b = c("y")))
- Substantively (not like below): Exhaustive and unambiguous

```
education_dictionary <- dictionary(list(higher_education=c("universit*","higher education","professor*"),school_ed
ucation=c("school*","teache*")))
education <- dfm_lookup(dfm_debate,education_dictionary)
education %>% textstat_frequency()
```

```
##            feature frequency rank docfreq group
## 1 school_education        12    1       8   all
## 2 higher_education         2    2       1   all
```

Dictionary creation: Best practice

1. Creating a top-features list
   - From data
2. Deductively selecting features
   - Which of these explain the theoretical concept well (unambiguously)
3. Manual validation
   - Human coders determine whether feature should be included or not



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(Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries

Ashley Muddiman, Shannon C. McGregor & Natalie Jomini Stroud

06

Selecting texts to work with

Choosing your texts

Tomorrow you will work on your own projects. It is important that you choose a text to work with that is available and interesting for you. Think about three questions:

1 Fit

How could this text represent an answer to a research question you find interesting?

2 Availability

Can you get access to the texts you need to answer your research questions?

3 Format

Can you use the format directly, or do you need to make it workable first (PDFs, scans, web pages …)?

More: Grimmer, Roberts & Stewart (2022), Ch. 3–4 (Selecting Documents)

Text data that I can give you

Tomorrow you can either work with data I give you or bring your own data. Here is what I can give you:

- **UN General Debates:** countries' annual UN General Assembly speeches, 1970–2015, 7,507 speeches. Variables: session, year, country, text.
- **Brexit Bill:** House of Commons debates on the EU Withdrawal Bill, Sep 2017–Jun 2018, 4,437 speeches. Variables: date, speaker, party, word count, text.
- **Eurydice:** EU descriptions of 43 European education systems, collected around 2022, 4,465 chapters. Variables: country, chapter, text, url.
- **US Presidential Debate 2020:** first Trump–Biden debate, 29 Sep 2020, 789 speaking turns. Variables: speaker, minute, text.
- … more. Ask me if you need help!

Supervised Project Work: Asking a Research Question and Selecting Data

Individually think about what you would like to analyse tomorrow. Think first about a research question and then about possible texts to work with.

Answer the three questions for the texts you are interested in. I come by to think it through and brainstorm with you. Raise your hand if you have an immediate question

1. Fit: How could this text represent an answer to your research question?
2. Availability: Can you get the texts you need?
3. Format: Can you use it directly, or does it need to be made workable first?

Tomorrow afternoon you can work with your own texts. Have them ready on your laptop.